



Unlocking Infinite Possibilities

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TABLE OF CONTENTS

Section	Title
Cover Page	Student Details
Preliminary Figures	Figures 1–8: Network Scan Results (Local Network)
Sample Task 1	Kenya Revenue Authority (KRA) — OSINT Reconnaissance
	Company Background
	Tools Used (WHOIS, NSLOOKUP, Wappalyzer, crt.sh, Have I Been Pwned, Maltego)
	Figure 8 – NSLOOKUP / DNS Resolution
	Figure 9 – Domain Information
	Figure 10 – Registrar Information
	Figure 11 – Open Ports Scan & Services
	Figure 12 – Wappalyzer Technology Stack
	Figure 13 – Certificate Transparency Records (crt.sh)
	Figure 14 – Intelligence X Search Results
	Figure 14 – Have I Been Pwned Results
	Figure 15 – Nmap ACK Scan / Firewall Detection
Task 2	Scan Specific Machine
	Figure 16 – Ipconfig Output
	Figure 17 – ARP Command
	Figure 18 – Nmap Service Version Scan (192.168.1.0)
	Figure 19 – Nmap OS Detection Scan

Section

Title

Figure 20 – Nmap SMB Enumeration Scan

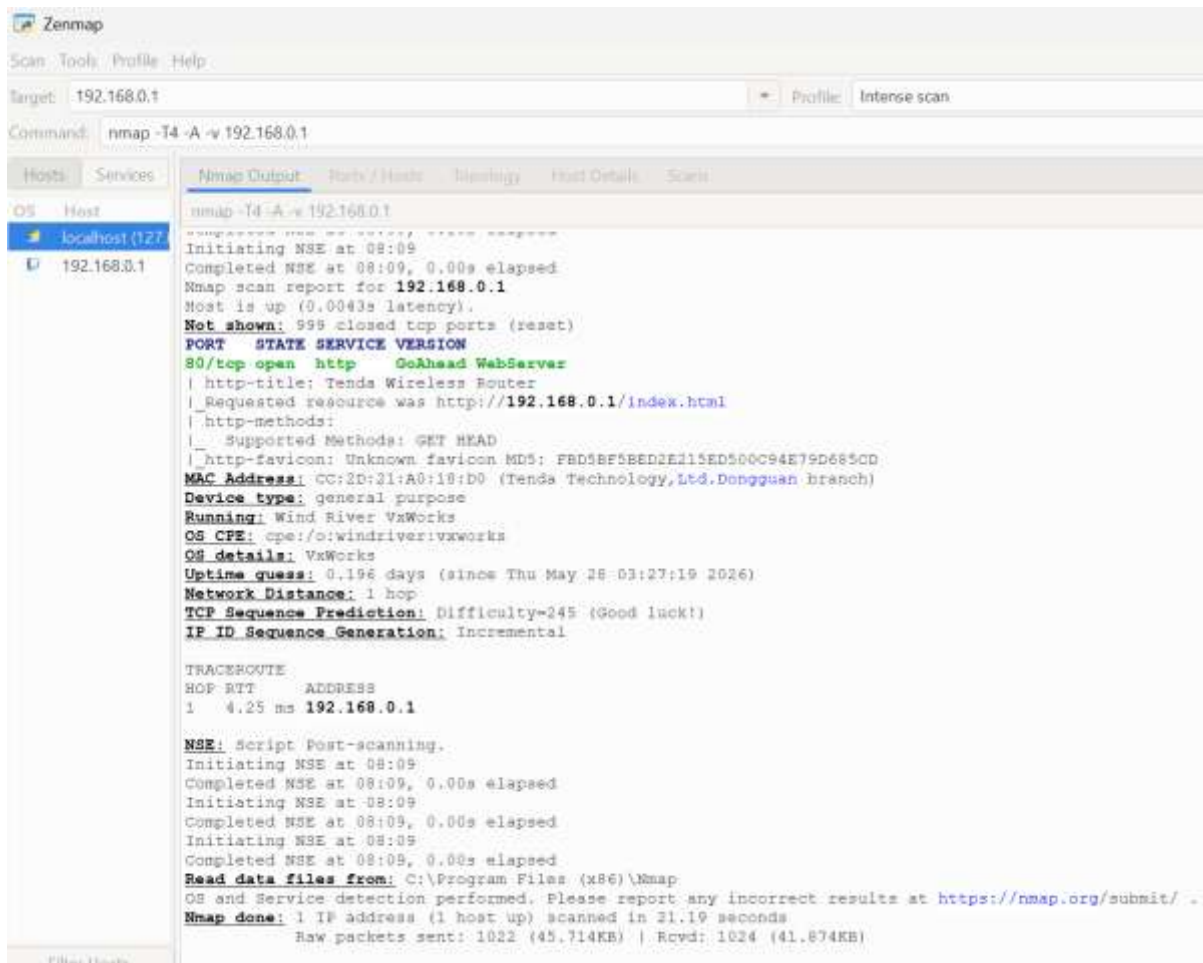


Fig 1: Nmap scan of router (192.168.0.1) showing open HTTP service and device detail

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\user> nmap -sn 192.168.1.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:15 +0300
Nmap scan report for 192.168.1.1
Host is up (0.011s latency).
Nmap scan report for 192.168.1.20
Host is up (0.11s latency).
Nmap scan report for 192.168.1.21
Host is up (0.11s latency).
Nmap scan report for 192.168.1.44
Host is up (0.040s latency).
Nmap scan report for 192.168.1.85
Host is up (0.038s latency).
Nmap scan report for 192.168.1.107
Host is up (0.050s latency).
Nmap scan report for 192.168.1.175
Host is up (0.21s latency).
Nmap scan report for 192.168.1.186
Host is up (0.010s latency).
Nmap scan report for 192.168.1.209
Host is up (0.019s latency).
Nmap scan report for 192.168.1.236
Host is up (0.029s latency).
Nmap scan report for 192.168.1.237
Host is up (0.028s latency).
Nmap scan report for 192.168.1.243
Host is up (0.023s latency).
Nmap scan report for 192.168.1.244
Host is up (0.023s latency).
Nmap scan report for 192.168.1.250
Host is up (0.026s latency).
```

Fig 2: Nmap host discovery scan showing active devices on the local network (192.168.1.0/24)

```

Link-local IPv6 Address . . . . . : fe80::c33f:3d4c:c42d:2d0c%20
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::992a:9000:2c24:c88a%21
    IPv4 Address. . . . . : 192.168.0.102
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter vEthernet (Default Switch):

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::7744:bc8a:c5ab:b586%22
    IPv4 Address. . . . . : 172.24.160.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . :

```

Fig 3: Windows IP configuration showing local network details and active network interfaces

```

PS C:\Users\user> nmap 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:25 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.20 seconds
PS C:\Users\user> nmap 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:25 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0067s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)

Nmap done: 1 IP address (1 host up) scanned in 1.91 seconds
PS C:\Users\user> |

```

Fig 4: Nmap host discovery scan executed in Windows PowerShell showing active devices within the local subnet and initial network reconnaissance results

```

Nmap done: 1 IP address (1 host up) scanned in 1.91 seconds
PS C:\Users\user> nmap -sV 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:28 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.31 seconds
PS C:\Users\user> nmap -sV 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:28 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0075s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      GoAhead WebServer
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.81 seconds
PS C:\Users\user> |

```

Fig 5: Nmap service version detection scan in Windows PowerShell identifying the specific web server application running on port 80 of the active target.

```

PS C:\Users\user> nmap -A 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:30 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.44 seconds
PS C:\Users\user> nmap -A 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:30 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0035s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http      GoAhead WebServer
|_ http-title: Tenda Wireless Router
|_ _Requested resource was http://192.168.0.1/index.html
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)
Device type: general purpose
Running: Wind River VxWorks
OS CPE: cpe:/o:windriver:vxworks
OS details: VxWorks
Network Distance: 1 hop

TRACEROUTE
HOP RTT      ADDRESS
1   3.49 ms  192.168.0.1

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.42 seconds
PS C:\Users\user> |

```

Fig 6: Nmap aggressive scan (-A) output showing operating system detection and HTTP script results for the target host.

```

PS C:\Users\user> nmap --script vuln 192.168.1.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:32 +0300
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 15.98 seconds
PS C:\Users\user> nmap --script vuln 192.168.0.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-28 08:33 +0300
Nmap scan report for 192.168.0.1
Host is up (0.0052s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-csrf: Couldn't find any CSRF vulnerabilities.
| http-fileupload-exploiter:
|
|_ Couldn't find a file-type field.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
MAC Address: CC:2D:21:A0:18:D0 (Tenda Technology,Ltd.Dongguan branch)

Nmap done: 1 IP address (1 host up) scanned in 235.83 seconds
PS C:\Users\user>

```

Fig 7: Nmap vulnerability scan (--script vuln) output displaying automated security assessment results for the open HTTP service on the target host.

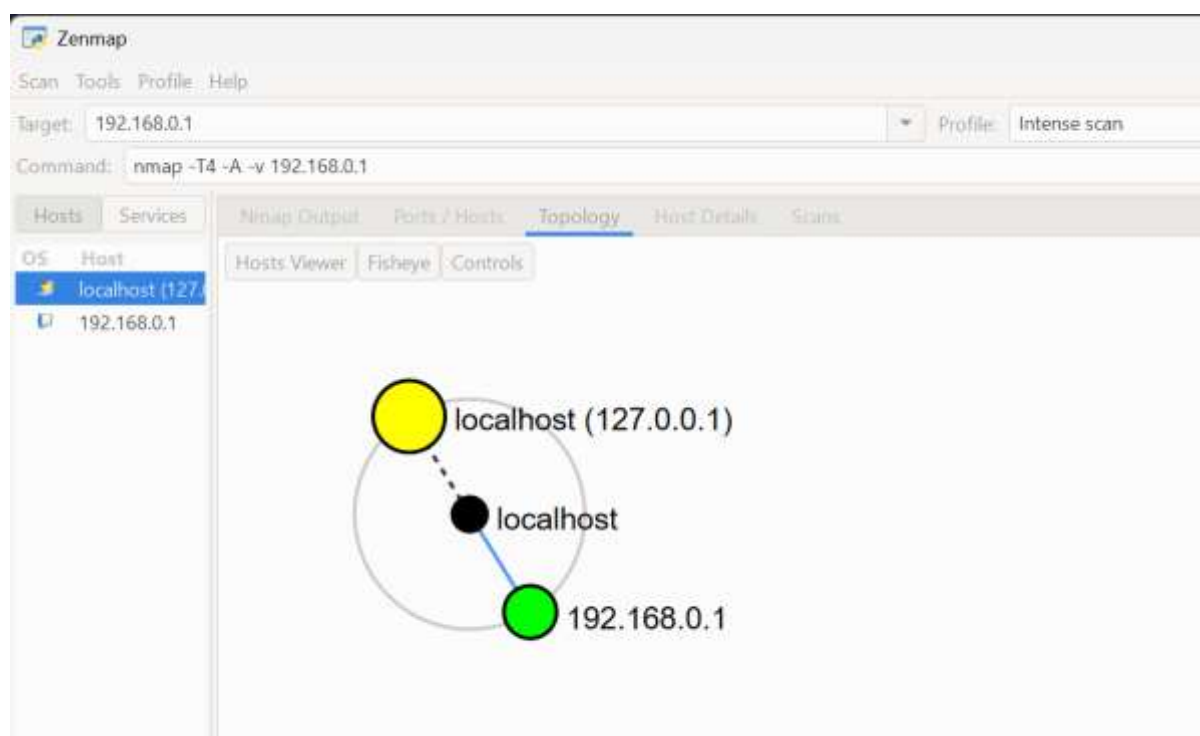


Fig 8: Zenmap graphical user interface displaying the network topology and node relationship between the local host and the target IP address.

SAMPLE TASK 1

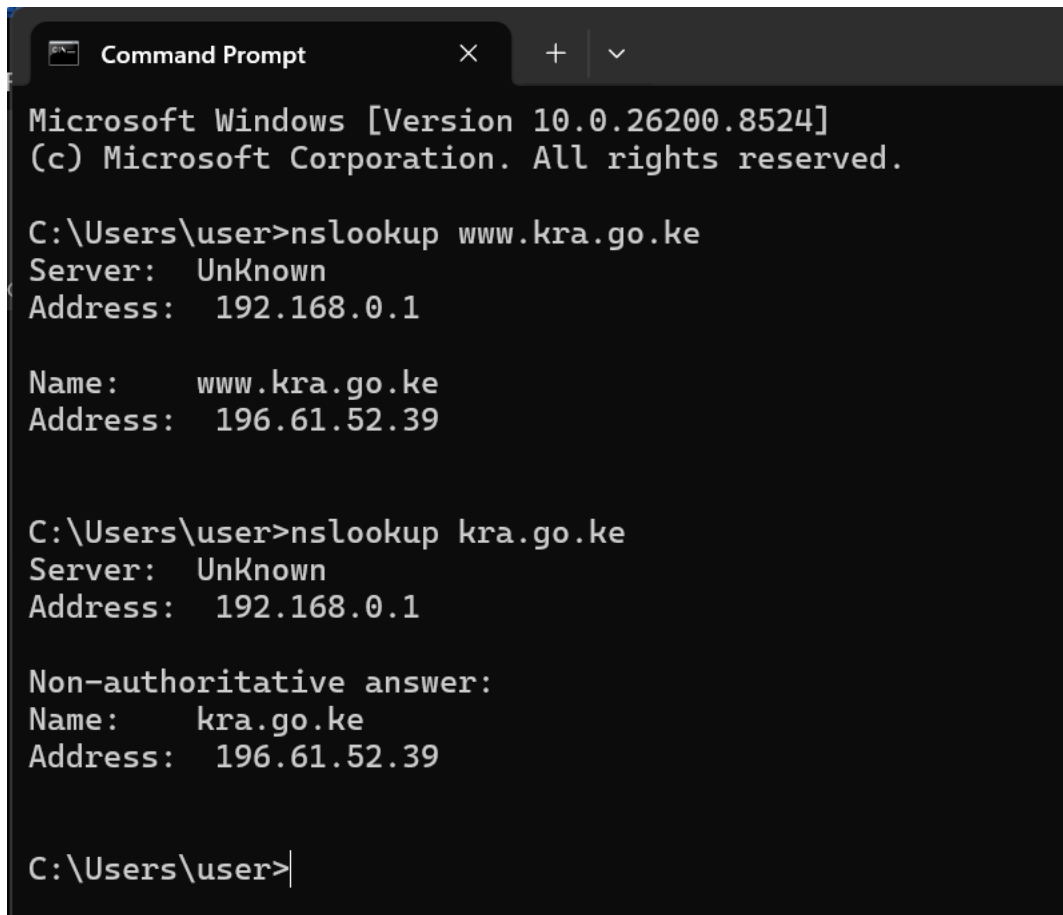
Company Background

Kenya Revenue Authorities

[Home - KRA](#)

Tools used

- WHOIS
- NSLOOKUP
- Wappalyzer
- crt.sh
- Have I Been Pwned
- Maltego



```
Command Prompt
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>nslookup www.kra.go.ke
Server: UnKnown
Address: 192.168.0.1

Name: www.kra.go.ke
Address: 196.61.52.39

C:\Users\user>nslookup kra.go.ke
Server: UnKnown
Address: 192.168.0.1

Non-authoritative answer:
Name: kra.go.ke
Address: 196.61.52.39

C:\Users\user>
```

Figure 8: Nslookup results showing DNS resolution and IP address information for the KRA domain (kra.go.ke).

kra.go.ke

Updated 1 day ago 

Domain Information

Domain:	kra.go.ke
Registered On:	2003-02-12
Expires On:	2026-12-31
Updated On:	2026-01-02
Status:	active
Name Servers:	ns4.safaricombusiness.co.ke ns03.mtnbusiness.co.ke ns01.mtnbusiness.co.ke ns1.safaricombusiness.co.ke ns2.safaricombusiness.co.ke ns3.safaricombusiness.co.ke ns02.mtnbusiness.co.ke

Figure 9: Domain Information

Registrar Information

Registrar: ICT Authority

Raw Whois Data

Domain Name: kra.go.ke
Registry Domain ID: 731-Kenic
Updated Date: 2026-01-02T07:05:00Z
Creation Date: 2003-02-12T21:00:00Z
Registry Expiry Date: 2026-12-31T21:00:00Z
Registrar Registration Expiration Date: 2026-12-31T21:00:00Z
Registrar: ICT Authority
Registrar Street Address: Teleposta Towers, 12th Flr,
Kenyatta Ave., Nairobi, Kenya.

P.O. Box 27150 - 00100,
Nairobi, Kenya.
Registrar City: Nairobi
Registrar State: Kenya
Registrar Postal Code: 00100
Registrar Phone: +254 20 2211960
Registrar Email: **websupport**@icta.go.ke
Registrar Abuse Contact Email:
Registrar Abuse Contact Phone:
Domain Status: active <https://icann.org/epp#active>
Name Server: ns4.safaricombusiness.co.ke
Name Server: ns03.mtnbusiness.co.ke

Figure 10: Registrar Information

```

C:\Users\user>nmap -sV kra.go.ke
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-03 23:51 +0300
Nmap scan report for kra.go.ke (196.61.52.39)
Host is up (0.028s latency).
Other addresses for kra.go.ke (not scanned): 64:ff9b::c43d:3427
Not shown: 695 closed tcp ports (reset), 201 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
43/tcp    open  tcpwrapped
79/tcp    open  tcpwrapped
80/tcp    open  tcpwrapped
84/tcp    open  tcpwrapped
109/tcp   open  tcpwrapped
161/tcp   open  tcpwrapped
443/tcp   open  tcpwrapped
465/tcp   open  tcpwrapped
515/tcp   open  tcpwrapped
668/tcp   open  tcpwrapped
687/tcp   open  tcpwrapped
800/tcp   open  tcpwrapped
902/tcp   open  tcpwrapped
903/tcp   open  tcpwrapped
1000/tcp  open  tcpwrapped
1042/tcp  open  tcpwrapped
1043/tcp  open  tcpwrapped
1055/tcp  open  tcpwrapped
1058/tcp  open  tcpwrapped
1070/tcp  open  tcpwrapped
1079/tcp  open  tcpwrapped
1082/tcp  open  tcpwrapped
1085/tcp  open  tcpwrapped
1099/tcp  open  tcpwrapped
1106/tcp  open  tcpwrapped
1122/tcp  open  tcpwrapped
1124/tcp  open  tcpwrapped
1132/tcp  open  tcpwrapped
1149/tcp  open  tcpwrapped
1154/tcp  open  tcpwrapped
1163/tcp  open  tcpwrapped
1174/tcp  open  tcpwrapped
1234/tcp  open  tcpwrapped
1311/tcp  open  tcpwrapped
1583/tcp  open  tcpwrapped
1594/tcp  open  tcpwrapped
1641/tcp  open  tcpwrapped
1688/tcp  open  tcpwrapped
1721/tcp  open  tcpwrapped

```

Figure 11: Open ports scan and services

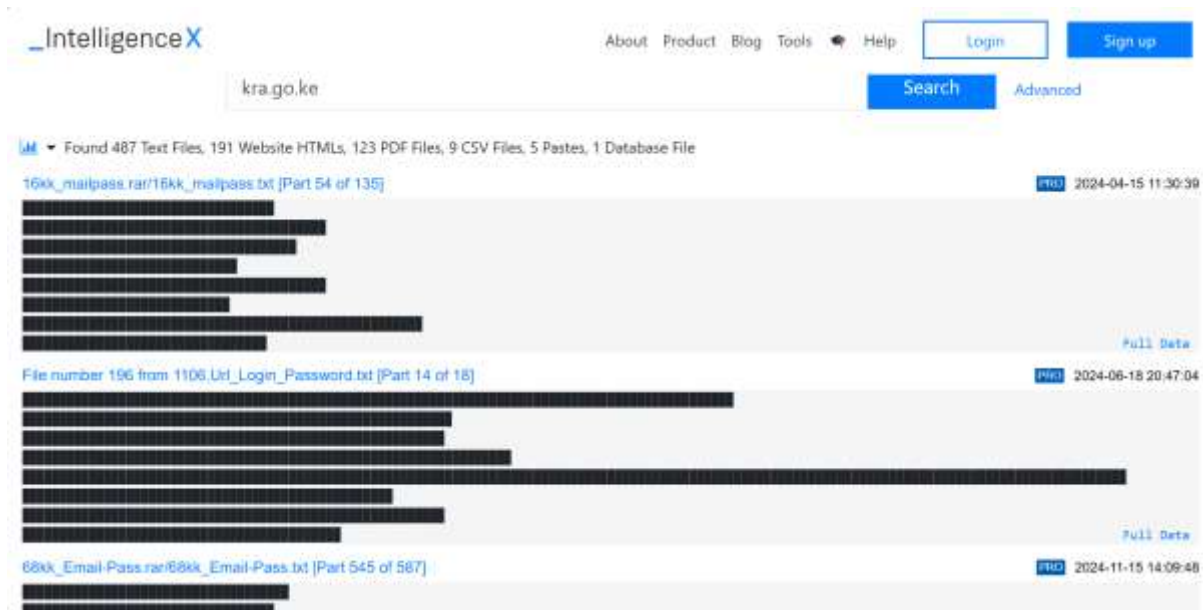


Figure 14: Intelligence X search results for kra.go.ke revealing 487 text files, 191 HTML files, 123 PDF files, 9 CSV files, 5 pastes and 1 database file containing leaked email and password combinations associated with the domain.

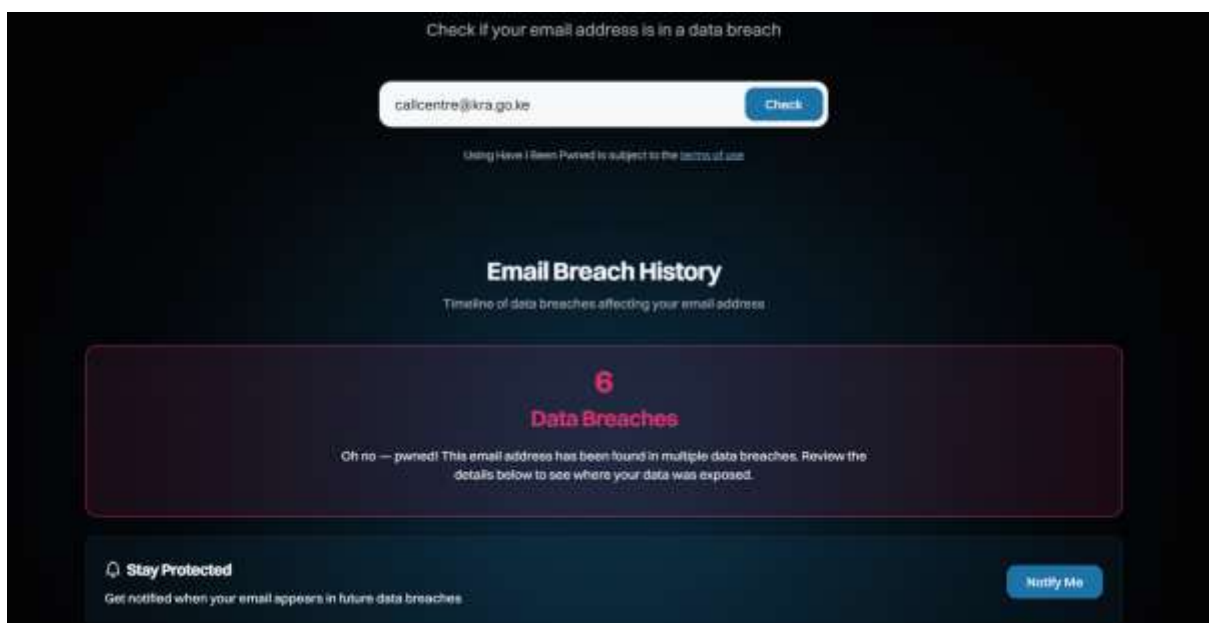


Figure 14: Have I Been Pwned results for callcentre@kra.go.ke confirming the email address has appeared in 6 data breaches, indicating significant credential exposure risk.

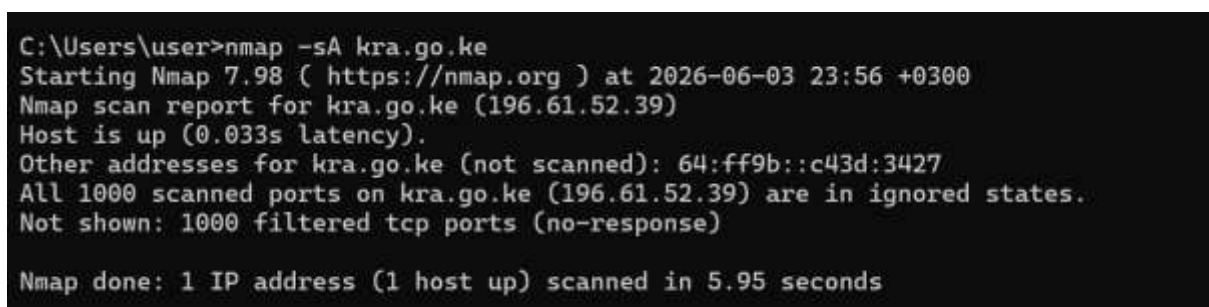


Figure 15: Nmap ACK scan on kra.go.ke (196.61.52.39) showing all 1000 scanned ports in a filtered state with no response, confirming the presence of a firewall protecting the organization's infrastructure.

Task 2: Scan Specific Machine

```
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::56e5:191f:dbbb:9a62%19
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Wireless LAN adapter WiFi:

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2c0f:fe38:2215:49e2:e668:ce2b:a202:8e8
Temporary IPv6 Address. . . . . : 2c0f:fe38:2215:49e2:ddc6:151f:c5ef:9c44
Link-local IPv6 Address . . . . . : fe80::992a:9000:2c24:c88a%20
IPv4 Address. . . . . : 10.49.135.34
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::c088:47ff:febe:b5ac%20
                             10.49.135.199

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Ethernet adapter vEthernet (Default Switch):

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::f308:fc21:5e83:b25%39
IPv4 Address. . . . . : 172.21.16.1
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . : 

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::1d4:47ed:4720:5713%63
IPv4 Address. . . . . : 172.19.96.1
Subnet Mask . . . . . : 255.255.240.0
```

Figure16:Ipconfig output displaying network adapter configurations and IP addresses of the host machine.

```

C:\Users\user>arp -anmap -p- 192.168.1.0

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

-a          Displays current ARP entries by interrogating the current
            protocol data. If inet_addr is specified, the IP and Physical
            addresses for only the specified computer are displayed. If
            more than one network interface uses ARP, entries for each ARP
            table are displayed.

-g          Same as -a.

-v          Displays current ARP entries in verbose mode. All invalid
            entries and entries on the loop-back interface will be shown.

inet_addr   Specifies an internet address.
-N if_addr  Displays the ARP entries for the network interface specified
            by if_addr.

-d          Deletes the host specified by inet_addr. inet_addr may be
            wildcarded with * to delete all hosts.

-s          Adds the host and associates the Internet address inet_addr
            with the Physical address eth_addr. The Physical address is
            given as 6 hexadecimal bytes separated by hyphens. The entry
            is permanent.

eth_addr    Specifies a physical address.
if_addr     If present, this specifies the Internet address of the
            interface whose address translation table should be modified.
            If not present, the first applicable interface will be used.

Example:
> arp -s 157.55.85.212 00-aa-00-62-c6-09 .... Adds a static entry.
> arp -a          .... Displays the arp table.

```

Figure17: Scan Ports, ARP command help page displayed due to incorrect syntax entry.

```

C:\Users\user>nmap -sV 192.168.1.0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-04 00:31 +0300
Nmap scan report for 192.168.1.0
Host is up (0.023s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE      SERVICE VERSION
53/tcp    filtered  domain

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.39 seconds

```

Figure18: Nmap service version scan on 192.168.1.0 revealing one open port — port 53 (DNS) in a filtered state, with 999 closed TCP ports detected.

```

C:\Users\user>nmap -O 192.168.1.0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-04 00:34 +0300
Nmap scan report for 192.168.1.0
Host is up (0.024s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE      SERVICE
53/tcp    filtered  domain
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: WAP/prnter/specialized/general purpose
Running (JUST GUESSING): D-link embedded (95%), HP embedded (87%), Linux 5.X (87%), STM Zurich Bluebottle (86%)
OS CPE: cpe:/o:dlink:dlink cpe:/o:hp:hp cpe:/o:linux:linux_kernel:5.11 cpe:/p:stmrzurich:bluebottle
Aggressive OS guesses: D-link D-324 wireless broadband router (93%), HP m241fdn printer (87%), Proxmox Virtual Environment (Linux 5.13) (87%), Bluebottle OS (86%), HP La
serJet 1240dtn printer (84%), HP LaserJet M printer (84%)
No exact OS matches for host (test conditions non-ideal).
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 28.93 seconds

```

Figure19: Nmap OS detection scan on 192.168.1.0 identifying the device type as a WAP/prnter with possible operating systems including D-Link embedded, HP embedded, and Linux 5.X.

```
C:\Users\user>nmap --script smb-enum-shares 192.168.1.0
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-04 00:37 +0300
Nmap scan report for 192.168.1.0
Host is up (0.024s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE      SERVICE
53/tcp    filtered  domain
10778/tcp filtered  unknown

Nmap done: 1 IP address (1 host up) scanned in 26.16 seconds
```

Figure20: Nmap SMB enumeration scan on 192.168.1.0 showing two filtered ports (53/TCP - DNS and 10778/TCP - unknown) with no shared folders accessible, indicating restricted network